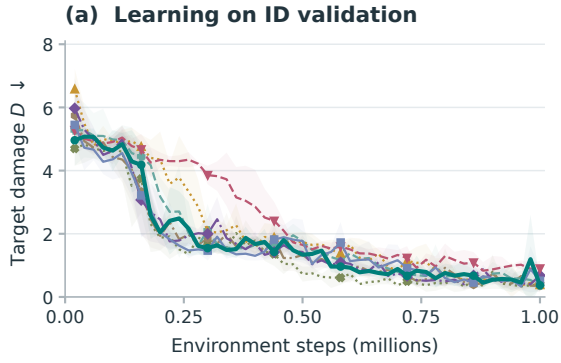
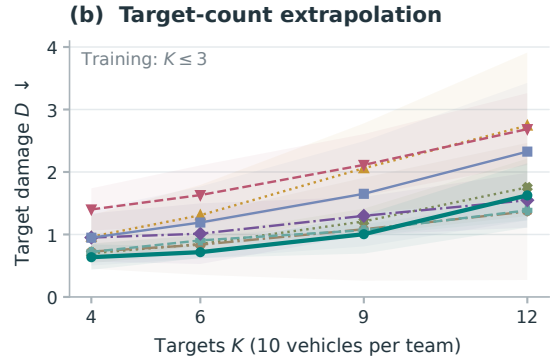
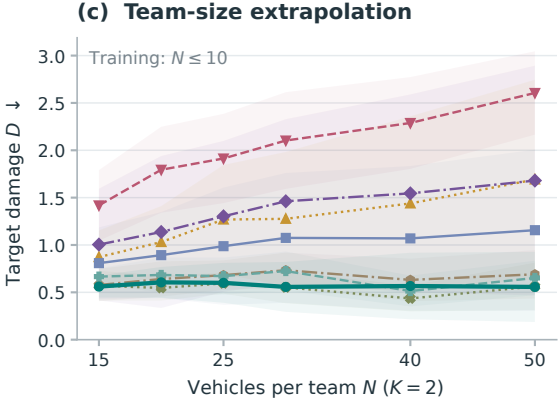
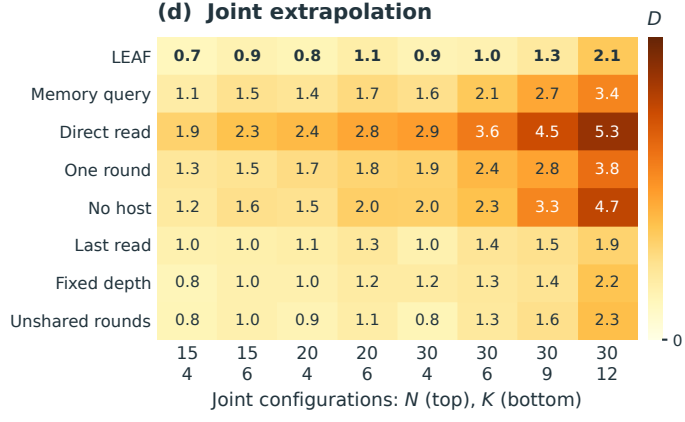

(a) Learning on ID validation
Target damage D ↓
Environment steps (millions)
0.00
0.25
0.50
0.75
1.00
0
2
4
6
8


(b) Target-count extrapolation
Training: K ≤ 3
Target damage D ↓
Targets K (10 vehicles per team)
4
6
9
12
0
1
2
3
4


(c) Team-size extrapolation
Training: N ≤ 10
Target damage D ↓
Vehicles per team N (K = 2)
15
25
40
50
0.0
0.5
1.0
1.5
2.0
2.5
3.0


(d) Joint extrapolation
D
0
LEAF 0.7 0.9 0.8 1.1 0.9 1.0 1.3 2.1
Memory query 1.1 1.5 1.4 1.7 1.6 2.1 2.7 3.4
Direct read 1.9 2.3 2.4 2.8 2.9 3.6 4.5 5.3
One round 1.3 1.5 1.7 1.8 1.9 2.4 2.8 3.8
No host 1.2 1.6 1.5 2.0 2.0 2.3 3.3 4.7
Last read 1.0 1.0 1.1 1.3 1.0 1.4 1.5 1.9
Fixed depth 0.8 1.0 1.0 1.2 1.2 1.3 1.4 2.2
Unshared rounds 0.8 1.0 0.9 1.1 0.8 1.3 1.6 2.3
15 15 20 20 30 30 30 30
4 6 4 6 4 6 9 12
Joint configurations: N (top), K (bottom)


LEAF
Memory query
Direct read
One round
No host
Last read
Fixed depth
Unshared rounds